

Payload dimensions:

- Weight: 160g
- Width: 80mm
- Height: 100mm
- Magnet position: (40, 40.1)

Robot dimensions:

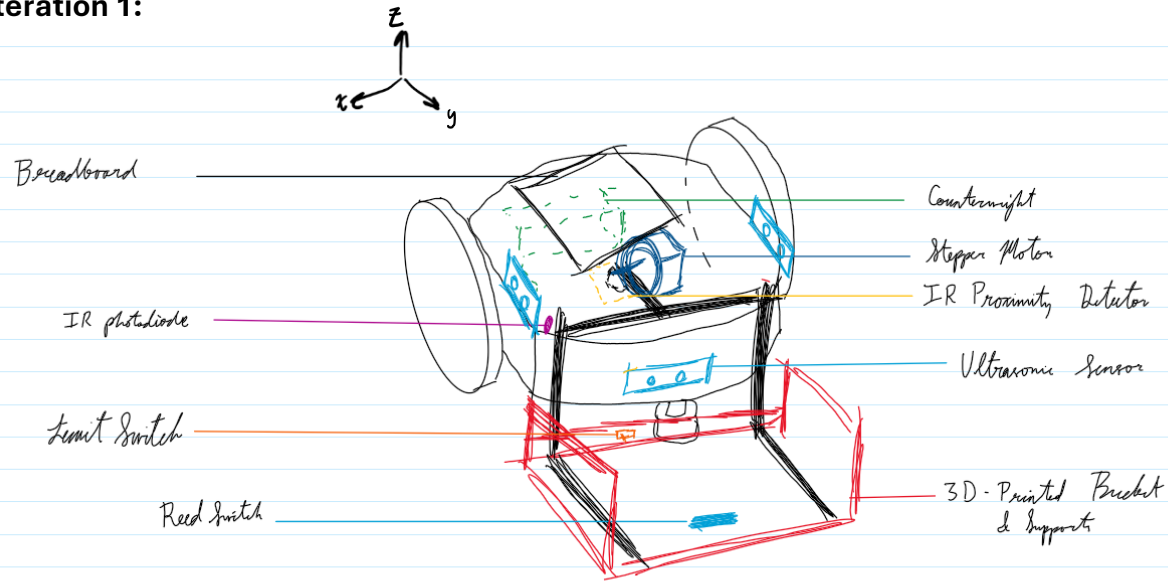
- Length:
- Width:

Component	GP Pin	Function returns
Motor Drivers	A: 6, 7, 8 B: 3, 4, 2	motor_a("direction", speed) motor_b("direction", speed)
Stepper Motor	10 11 12 13	
Servo Motor		
Reed Switch	20	reed_switch.value()
Limit Switch	10	button.value()
Ultrasonic Sensor	GP: (Echo, Trigger) -> Phys. 1 (F): 20, 21 -> 26, 27 2 (R): 18, 19 -> 24, 25 3 (L): 16, 17 -> 21, 22	<i>Variable:</i> "distance"
IR Proximity Detector	15	line_sen.value()
Gyroscope	2, 3	<i>Variables:</i> "imu.accel.xyz", "imu.gyro.xyz"
Photodiode	28	ir.read_u16()

Friday, March 14:

Created Gantt chart, assigned tasks, deadlines, and discussed dependencies. Discussed different solutions and sketched them. Designed a final sketch to base our prototype on. Discussed dimensions for the 3D prints as well as the flow of code for the components we plan to use.

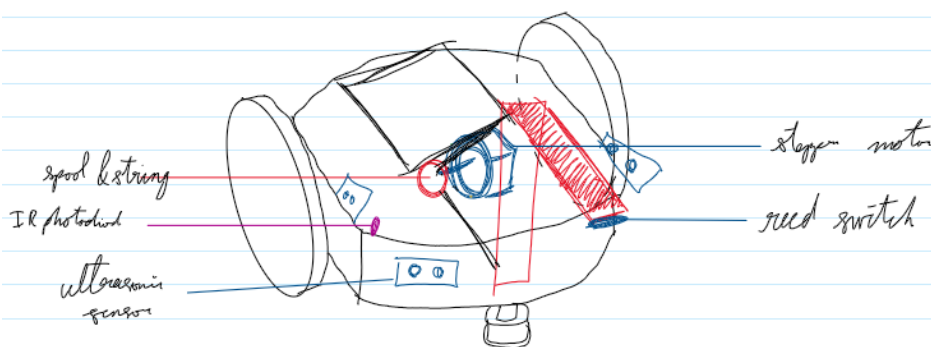
Iteration 1:



Tuesday, March 18:

Tested ultrasonic sensor, developed flowchart and code for electronic components. Improved payload lifting mechanism for increased consistency when picking up the payload.

Iteration 2:



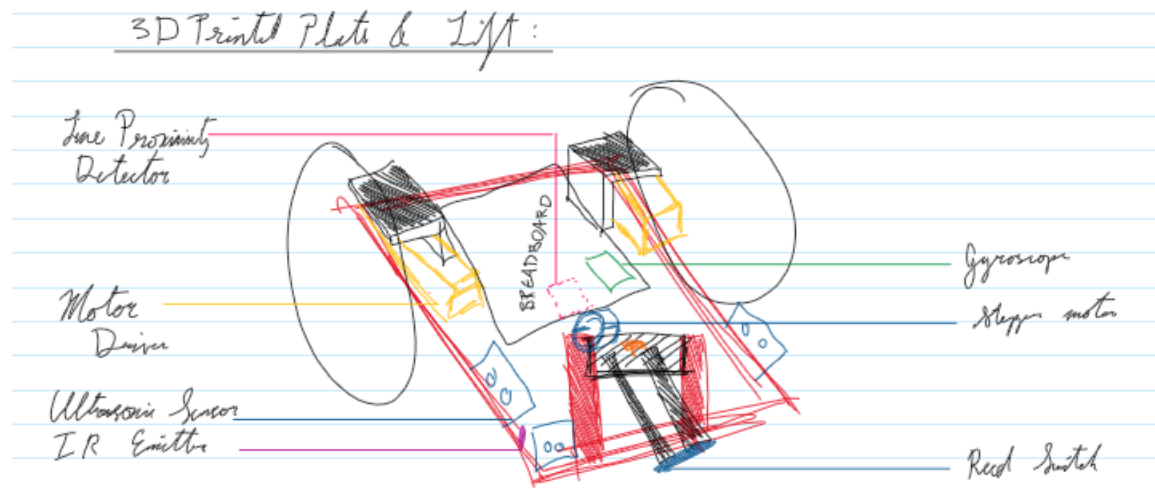
- Lifting mechanism's height controlled by a spool and string connected to the stepper motor. Robot will drive with the 3D-printed lever lowered so the reed switch

can detect the magnet at the proper height (33.2mm) then raise to (~80mm) when a payload is detected.

Friday, March 21:

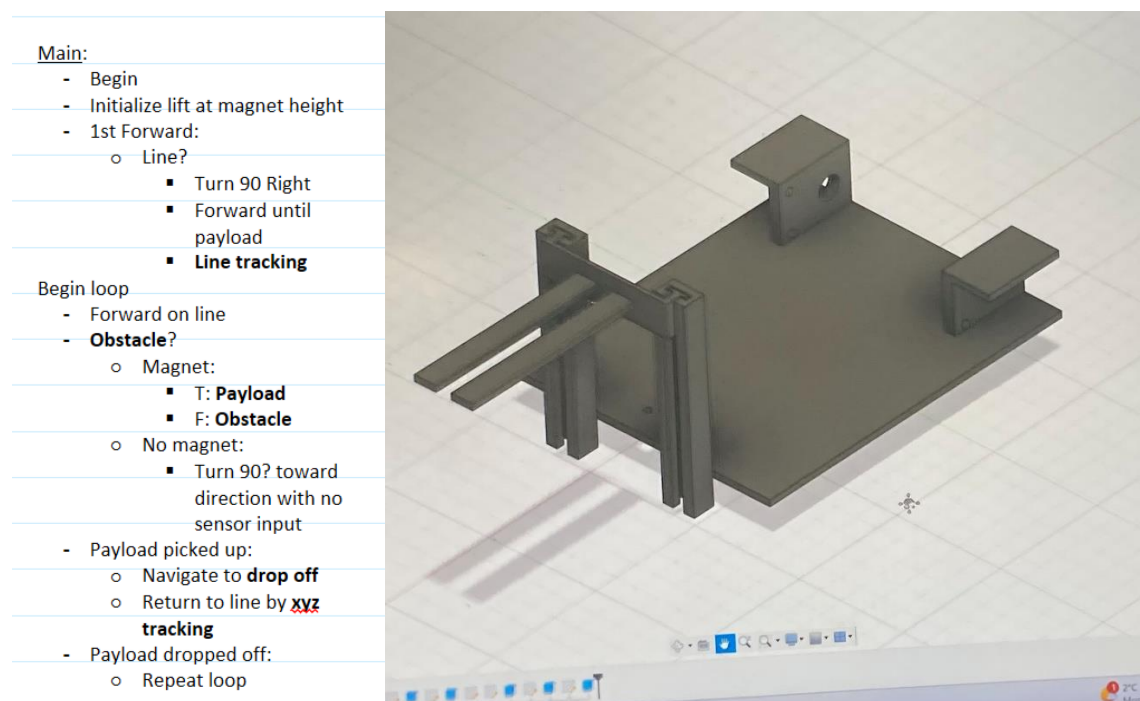
Finalized flowcharts, tested all components, and worked on 3D modelling/code.

Iteration 3:



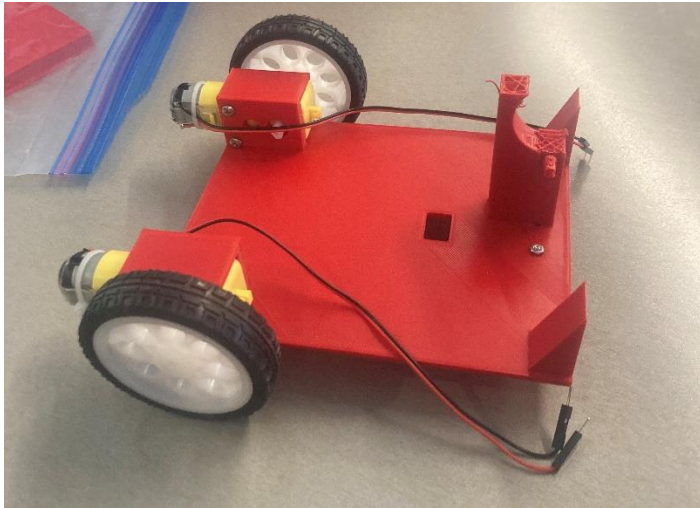
Moved limit switch to the front for obstacle detection and payload sensing. Obstacle detected if only limit switch on. Payload detected if both reed switch and limit switch on.

Basic Functionality Flowchart and modelling:



Thursday, March 27:

Tested code so far for motors, checked motor drivers with wheels and 3D printed model.



Friday, March 28:

Planned to revise the 3D model (**Iteration 4**):

- rails and wall extended to account for front ultrasonic sensor
- forklift extended
- tires moved up, caster ball moved back
- mounts for the ultrasonic sensors

Began to compile code onto a main code

Monday, March 31:

Tested and attached photodiode and reed switch for pre-competition check

Tuesday, April 1:

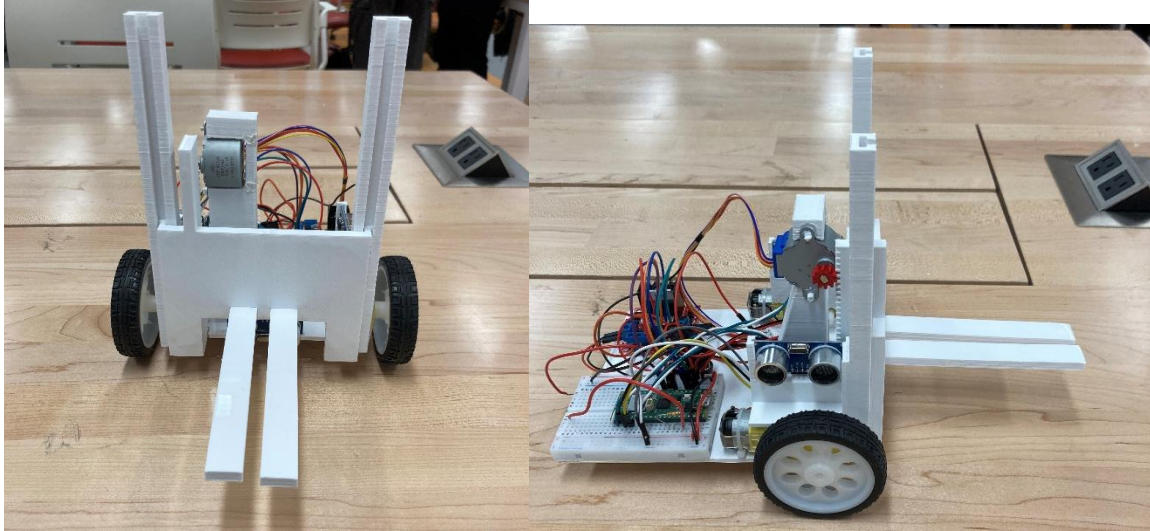
Confirmed pre-competition check functionalities. Outlined pitch and technical report and updated the presentation.

Friday, April 4:

Worked on code and wiring

Tuesday, April 8

Worked on obstacle code and focused on bonus tests, printed and put together the final prototype.



Wednesday, April 9

Finalized obstacle code and payload code for bonus tests.

Thursday, April 10

Finalized code and robot